

Savannah Tourism Database System Design - Extended Entity Relation Design

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I. Introduction:

This report details the systematic analysis and design of a relational database system for the publisher of the book "Glorious Savannah". The database design adheres strictly to two core requirements: **Extended Entity-Relationship (EER)** modeling for the conceptual design and the **systematic conversion of the EER model** into a final set of relations. This document is organized to present the Blocking Process, the resulting EER Diagram with all constraints and reasoning, the systematic Conversion of the EER model into relations, a list of attributes, and the Final List of Relation Schemas.

II. Blocking Process:

1. Each Book has an ISBN, title, publisher, edition number, year of publication, and a number of pages. Each Book has several Attractions. Each Book has several Businesses. Each Book has several Hotels. Each Book has several Restaurants.
2. Each Book has several Attractions. Each Attraction has an identification number. Each Attraction has several Emergency Agencies, several Savannah area attractions, several Tybee Island area attractions, several Historic Cemeteries, several Historic Churches, several Historic Squares, several Public Parks, several Events, several Art attractions, several Sports attractions, several Universities.
3. Each Attraction has several Emergency Agencies. Each Emergency Agency has an identification number, name, and phone number.
4. Each Attraction has several Savannah area attractions. Each Savannah area attraction has an identification number, name, location, phone number, website, and description.
5. Each Attraction has several Tybee Island area attractions. Each Tybee Island area attraction has an identification number, name, location, phone number, website, and description.
6. Each Attraction has several Historic Cemeteries. Each Historic Cemetery has an identification number, name, location, year established, type, and description.
7. Each Attraction has several Historic Churches. Each Historic Church has an identification number, name, phone number, location, denomination, website, and description.
8. Each Attraction has several Historic Squares. Each Historic Square has an identification number, name, a location, a designer, a design year, architecture style, and description.
9. Each Attraction has several Public Parks. Each Public Park has an identification number, name, location, municipality, and description.
10. Each Attraction has several Events. Each Calendar Event has an identification number, name, phone number, location, website, start date, and description.
11. Each Attraction has several Art attractions. Each Art attraction has an identification number, name, phone number, location, website, and type.

12. Each Attraction has several Sports attractions. Each Sports attraction has an identification number. Each Sports attraction has several golf courses. Each Sports attraction has several tennis courses. Each Sports attraction has several boat ramps. Each Sports attraction has several fishing piers.
13. Each Sports attraction has several golf courses. Each Golf Course has an identification number, name, phone number, location, website, and description.
14. Each Sports attraction has several tennis courses. Each Tennis Course has an identification number, name, phone number, location, and description.
15. Each Sports attraction has several boat ramps. Each Boating Ramp has an identification number, name and location.
16. Each Sports attraction has several fishing piers. Each Fishing Pier has an identification number, name and location.
17. Each University has an identification number, name, phone number, email address, location, website, type, and description. Each University has several colleges. Each University has several programs. Each University has several departments.
18. Each University has several colleges. Each college has an identification number and name.
19. Each University has several programs. Each program has an identification number and name.
20. Each University has several departments. Each department in a university has an identification number, name, and phone number.
21. Each Book has several Businesses. Each Business has an identification number. Each Business has several Medical Establishments. Each Business has several Pest Control Companies. Each Business has several Plumbers.
22. Each Business has several Medical Establishments. Each Medical Establishment has an identification number, name, a specialty, a description, services offered, a location, hours of operation, website(s), and phone number(s). Each Medical Establishment has several doctors.
23. Each Medical Establishment has several doctors. Each Doctor has an identification number and name. Each Doctor has several degrees. Each degree has a name and level.
24. Each Business has several Pest Control Companies. Each Pest Control Company has an identification number, name, location, phone number, operating hours, specialty, payment method accepted, and description.
25. Each Business has several Plumbers. Each Plumber has an identification number, name, location, phone number, operating hours, specialty, payment method accepted, and description.
26. Each Book has several Hotels. Each Hotel has an identification number, name, phone number, location, website, price range, booking restrictions, amenities, payment method accepted, historic area, description, and cancellation restrictions,

27. Each Book has several Restaurants. Each Restaurant has an identification number, name, phone number, location, website, food specialty, may or may not accept reservations, and description.

II.1 Details for Each Block:

1. Each Book has an ISBN, title, publisher, edition number, year of publication, and a number of pages. Each Book has several Attractions. Each Book has several Businesses. Each Book has several Hotels. Each Book has several Restaurants.
 - a. Name of the entity: Book
 - b. The nature of the entity (i.e., superclass, subclass, regular, or weak entity):
Superclass, Regular
 - c. Name(s) of the entity's superclass(es), if applicable: N/A
 - d. Name(s) of its subclass(es), if applicable: Attractions, Businesses, Hotels, Restaurants
 - e. Its attributes: ISBN (ISBN), Title (Title), Publisher (Publisher), Edition number (Edition#), Year of publication (PubYear), Number of pages (NumPgs)
 - f. Its relationship(s) with other entities, if applicable: Attractions, Businesses, Hotels, and Restaurants are **sections of** Book
2. Each Book has several Attractions. Each Attraction has an identification number. Each Attraction has several Emergency Agencies, several Savannah area attractions, several Tybee Island area attractions, several Historic Cemeteries, several Historic Churches, several Historic Squares, several Public Parks, several Events, several Art attractions, several Sports attractions, several Universities.
 - a. Name of the entity: Attractions
 - b. The nature of the entity (i.e., superclass, subclass, regular, or weak entity):
Subclass, Regular
 - c. Name(s) of the entity's superclass(es), if applicable: Book
 - d. Name(s) of its subclass(es), if applicable: Emergency agencies, Savannah area attraction, Tybee Island area attractions, Historic Cemeteries, Historic Churches, Historic Squares, Parks, Events, Art attractions, Sports attractions, Universities
 - e. Its attributes: ID number (Att#)
 - f. Its relationship(s) with other entities, if applicable: Is section of Book
3. Each Attraction has several Emergency Agencies. Each Emergency Agency has an identification number, name, and phone number.
 - a. Name of the entity: Emergency Agency
 - b. The nature of the entity (i.e., superclass, subclass, regular, or weak entity):
Subclass, Regular
 - c. Name(s) of the entity's superclass(es), if applicable: Attractions
 - d. Name(s) of its subclass(es), if applicable: N/A

- e. Its attributes: ID number (Emergency#), Name (Emergency_Name), Phone number (Emergency_Phone)
 - f. Its relationship(s) with other entities, if applicable: Type of Attraction
- 4. Each Attraction has several Savannah area attractions. Each Savannah area attraction has an identification number, name, location, phone number, website, and description.
 - a. Name of the entity: Savannah area attractions
 - b. The nature of the entity (i.e., superclass, subclass, regular, or weak entity): Subclass, Regular
 - c. Name(s) of the entity's superclass(es), if applicable: Attractions
 - d. Name(s) of its subclass(es), if applicable: N/A
 - e. Its attributes: ID number (SavAtt#), Name (SavAtt_Name), Location (SavAtt_Loc), Phone number (SavAtt_Phone), Website (SavAtt_Web), Description (SavAtt_Descript)
 - f. Its relationship(s) with other entities, if applicable: Type of Attraction
- 5. Each Attraction has several Tybee Island area attractions. Each Tybee Island area attraction has an identification number, name, location, phone number, website, and description.
 - a. Name of the entity: Tybee Island area attractions
 - b. The nature of the entity (i.e., superclass, subclass, regular, or weak entity): Subclass, Regular
 - c. Name(s) of the entity's superclass(es), if applicable: Attractions
 - d. Name(s) of its subclass(es), if applicable: N/A
 - e. Its attributes: ID number (TybAtt#), Name (TybAtt_Name), Location (TybAtt_Loc), Phone number (TybAtt_Phone), Website (TybAtt_Web), Description (TybAtt_Descript)
 - f. Its relationship(s) with other entities, if applicable: Type of Attraction
- 6. Each Attraction has several Historic Cemeteries. Each Historic Cemetery has an identification number, name, location, year established, type, and description.
 - a. Name of the entity: Historic Cemeteries
 - b. The nature of the entity (i.e., superclass, subclass, regular, or weak entity): Subclass, Regular
 - c. Name(s) of the entity's superclass(es), if applicable: Attractions
 - d. Name(s) of its subclass(es), if applicable: N/A
 - e. Its attributes: ID number (Cemetery#), Name (Cemetery_Name), Location (Cemetery_Loc), Year established (Cemetery_EstYear), Type (Cemetery_Type), Description (Cemetery_Descript)
 - f. Its relationship(s) with other entities, if applicable: Type of Attraction
- 7. Each Attraction has several Historic Churches. Each Historic Church has an identification number, name, phone number, location, denomination, website, and description.
 - a. Name of the entity: Historic Churches

- b. The nature of the entity (i.e., superclass, subclass, regular, or weak entity): Subclass, Regular
 - c. Name(s) of the entity's superclass(es), if applicable: Attractions
 - d. Name(s) of its subclass(es), if applicable: N/A
 - e. Its attributes: ID number (Church#), Name (Church_Name), Phone number (Church_Phone), Location (Church_Loc), Denomination (Church_Denom), Website (Church_Web), Description (Church_Descript)
 - f. Its relationship(s) with other entities, if applicable: Type of Attraction
- 8. Each Attraction has several Historic Squares. Each Historic Square has an identification number, name, a location, a designer, a design year, architecture style, and description.
 - a. Name of the entity: Historic Squares
 - b. The nature of the entity (i.e., superclass, subclass, regular, or weak entity): Subclass, Regular
 - c. Name(s) of the entity's superclass(es), if applicable: Attractions
 - d. Name(s) of its subclass(es), if applicable: N/A
 - e. Its attributes: ID number (Square#), Name (Square_Name), Location (Square_Loc), Designer (Square_Designer), Design Year (Square_DesignYear), Architecture style (Square_ArchStyle), Description (Square_Descript)
 - f. Its relationship(s) with other entities, if applicable: Type of Attraction
- 9. Each Attraction has several Public Parks. Each Public Park has an identification number, name, location, municipality, and description.
 - a. Name of the entity: Park
 - b. The nature of the entity (i.e., superclass, subclass, regular, or weak entity): Subclass, Regular
 - c. Name(s) of the entity's superclass(es), if applicable: Attractions
 - d. Name(s) of its subclass(es), if applicable: N/A
 - e. Its attributes: ID number (Park#), Name (Park_Name), Location (Park_Loc), Municipality (Park_Munic), Description (Park_Descript)
 - f. Its relationship(s) with other entities, if applicable: Type of Attraction
- 10. Each Attraction has several Events. Each Calendar Event has an identification number, name, phone number, location, website, start date, and description.
 - a. Name of the entity: Calendar Events
 - b. The nature of the entity (i.e., superclass, subclass, regular, or weak entity): Subclass, Regular
 - c. Name(s) of the entity's superclass(es), if applicable: Attractions
 - d. Name(s) of its subclass(es), if applicable: N/A
 - e. Its attributes: ID number (CalEvent#), Name (CalEvent_Name), Phone number (CalEvent_Phone), Location (CalEvent_Loc), Website (CalEvent_Web), Start date (CalEvent_StartDate), Description (CalEvent_Descript)
 - f. Its relationship(s) with other entities, if applicable: Type of Attraction

11. Each Attraction has several Art attractions. Each Art attraction has an identification number, name, phone number, location, website, and type.
 - a. Name of the entity: Art Attraction
 - b. The nature of the entity (i.e., superclass, subclass, regular, or weak entity): Subclass, Regular
 - c. Name(s) of the entity's superclass(es), if applicable: Attractions
 - d. Name(s) of its subclass(es), if applicable: N/A
 - e. Its attributes: ID number (Art#), Name (Art_Name), Phone number (Art_Phone), Location (Art_Loc), Website (Art_Web), Type (Art_Type)
 - f. Its relationship(s) with other entities, if applicable: Type of Attraction
12. Each Attraction has several Sports attractions. Each Sports attraction has an identification number. Each Sports attraction has several golf courses. Each Sports attraction has several tennis courses. Each Sports attraction has several boat ramps. Each Sports attraction has several fishing piers.
 - a. Name of the entity: Sports
 - b. The nature of the entity (i.e., superclass, subclass, regular, or weak entity): Superclass & Subclass; Regular
 - c. Name(s) of the entity's superclass(es), if applicable: Attractions
 - d. Name(s) of its subclass(es), if applicable: Golf courses, Tennis courts, Boat ramps, Fishing piers
 - e. Its attributes: ID number (Sports#)
 - f. Its relationship(s) with other entities, if applicable: Type of Attraction
13. Each Sports attraction has several golf courses. Each Golf Course has an identification number, name, phone number, location, website, and description.
 - a. Name of the entity: Golf Course
 - b. The nature of the entity (i.e., superclass, subclass, regular, or weak entity): Subclass, Regular
 - c. Name(s) of the entity's superclass(es), if applicable: Sports
 - d. Name(s) of its subclass(es), if applicable: N/A
 - e. Its attributes: ID number (Golf#), Name (Golf_Name), Phone number (Golf_Phone), Location (Golf_Loc), Website (Golf_Web), Description (Golf_Descript)
 - f. Its relationship(s) with other entities, if applicable: Type of Sports attraction
14. Each Sports attraction has several tennis courses. Each Tennis Course has an identification number, name, phone number, location, and description.
 - a. Name of the entity: Tennis Course
 - b. The nature of the entity (i.e., superclass, subclass, regular, or weak entity): Subclass, Regular
 - c. Name(s) of the entity's superclass(es), if applicable: Sports
 - d. Name(s) of its subclass(es), if applicable: N/A

- e. Its attributes: ID number (Tennis#), Name (Tennis_Name), Phone number (Tennis_Phone), Location (Tennis_Loc), Website (Tennis_Web), Description (Tennis_Descript)
 - f. Its relationship(s) with other entities, if applicable: Type of Sports attraction
15. Each Sports attraction has several boat ramps. Each Boating Ramp has an identification number, name and location.
- a. Name of the entity: Boat Ramp
 - b. The nature of the entity (i.e., superclass, subclass, regular, or weak entity): Subclass, Regular
 - c. Name(s) of the entity's superclass(es), if applicable: Sports
 - d. Name(s) of its subclass(es), if applicable: N/A
 - e. Its attributes: ID number (Ramp#), Name (Ramp_Name), Location (Ramp_Loc)
 - f. Its relationship(s) with other entities, if applicable: Type of Sports attraction
16. Each Sports attraction has several fishing piers. Each Fishing Pier has an identification number, name and location.
- a. Name of the entity: Fishing Pier
 - b. The nature of the entity (i.e., superclass, subclass, regular, or weak entity): Subclass, Regular
 - c. Name(s) of the entity's superclass(es), if applicable: Sports
 - d. Name(s) of its subclass(es), if applicable: N/A
 - e. Its attributes: ID number (Pier#), Name (Pier_Name), Location (Pier_Loc)
 - f. Its relationship(s) with other entities, if applicable: Type of Sports attraction
17. Each University has an identification number, name, phone number, email address, location, website, type, and description. Each University has several colleges. Each University has several programs. Each University has several departments.
- a. Name of the entity: University
 - b. The nature of the entity (i.e., superclass, subclass, regular, or weak entity): Superclass & Subclass; Regular
 - c. Name(s) of the entity's superclass(es), if applicable: Attractions
 - d. Name(s) of its subclass(es), if applicable: Colleges, Programs, Departments
 - e. Its attributes: ID number (Uni#), Name (Uni_Name), Phone number (Uni_Phone), Email address (Uni_Email), Location (Uni_Loc), Website (Uni_Web), Type (Uni_Type), Description (Uni_Descript)
 - f. Its relationship(s) with other entities, if applicable: Type of Attraction
18. Each University has several colleges. Each college has an identification number and name.
- a. Name of the entity: University colleges
 - b. The nature of the entity (i.e., superclass, subclass, regular, or weak entity): Subclass, Regular
 - c. Name(s) of the entity's superclass(es), if applicable: University

- d. Name(s) of its subclass(es), if applicable: N/A
 - e. Its attributes: ID number (UniCollege#), Name (UniCollege_Name)
 - f. Its relationship(s) with other entities, if applicable: Part of University
19. Each University has several programs. Each program has an identification number and name.
- a. Name of the entity: University programs
 - b. The nature of the entity (i.e., superclass, subclass, regular, or weak entity): Subclass, Regular
 - c. Name(s) of the entity's superclass(es), if applicable: University
 - d. Name(s) of its subclass(es), if applicable: N/A
 - e. Its attributes: ID number (UniProg#), Name (UniProg_Name)
 - f. Its relationship(s) with other entities, if applicable: Part of University
20. Each University has several departments. Each department in a university has an identification number, name, and phone number.
- a. Name of the entity: University departments
 - b. The nature of the entity (i.e., superclass, subclass, regular, or weak entity): Subclass, Regular
 - c. Name(s) of the entity's superclass(es), if applicable: University
 - d. Name(s) of its subclass(es), if applicable: N/A
 - e. Its attributes: ID number (UniDept#), Name (UniDept_Name), Phone number (UniDept_Phone)
 - f. Its relationship(s) with other entities, if applicable: Part of University
21. Each Book has several Businesses. Each Business has an identification number. Each Business has several Medical Establishments. Each Business has several Pest Control Companies. Each Business has several Plumbers.
- a. Name of the entity: Businesses
 - b. The nature of the entity (i.e., superclass, subclass, regular, or weak entity): Subclass, Regular
 - c. Name(s) of the entity's superclass(es), if applicable: Book
 - d. Name(s) of its subclass(es), if applicable: Medical establishments, Pest control companies, Plumbers
 - e. Its attributes: ID number (BusID)
 - f. Its relationship(s) with other entities, if applicable: Is section of Book
22. Each Business has several Medical Establishments. Each Medical Establishment has an identification number, name, a specialty, a description, services offered, a location, hours of operation, website(s), and phone number(s). Each Medical Establishment has several doctors.
- a. Name of the entity: Medical Establishment
 - b. The nature of the entity (i.e., superclass, subclass, regular, or weak entity): Superclass & Subclass, Regular

- c. Name(s) of the entity's superclass(es), if applicable: Business
 - d. Name(s) of its subclass(es), if applicable: Doctors
 - e. Its attributes: ID number (Med#), Name (MedEst_Name), Specialty (MedEst_Specialty), Description (MedEst_Descript), Services offered (MedEst_Services), Location (MedEst_Loc), Hours of operation (MedEst_OpHours), Website (MedEst_Web), and Phone number (MedEst_Phone)
 - f. Its relationship(s) with other entities, if applicable: Type of Business
23. Each Medical Establishment has several doctors. Each Doctor has an identification number and name. Each Doctor has several degrees. Each degree has a name and level.
- a. Name of the entity: Doctors
 - b. The nature of the entity (i.e., superclass, subclass, regular, or weak entity): Subclass, Weak
 - c. Name(s) of the entity's superclass(es), if applicable: Medical Establishments
 - d. Name(s) of its subclass(es), if applicable: N/A
 - e. Its attributes: ID number (Doc#), Name (Doc_Name), Degrees (Doc_Degree); Composite Attributes w Doc_Degree: Name (Degree_Name), Level (Degree_Lvl)
 - f. Its relationship(s) with other entities, if applicable: Employed by Medical Establishment
24. Each Business has several Pest Control Companies. Each Pest Control Company has an identification number, name, location, phone number, operating hours, specialty, payment method accepted, and description.
- a. Name of the entity: Pest Control Companies
 - b. The nature of the entity (i.e., superclass, subclass, regular, or weak entity): Subclass, Regular
 - c. Name(s) of the entity's superclass(es), if applicable: Businesses
 - d. Name(s) of its subclass(es), if applicable: N/A
 - e. Its attributes: ID number (Pest#), Name (Pest_Name), Location (Pest_Loc), Phone number (Pest_Phone), Operating hours (Pest_OpHours), Specialty (Pest_Specialty), Accepted payment methods (Pest_AccPmt), Description (Pest_Descript)
 - f. Its relationship(s) with other entities, if applicable: Type of Business
25. Each Business has several Plumbers. Each Plumber has an identification number, name, location, phone number, operating hours, specialty, payment method accepted, and description.
- a. Name of the entity: Plumber
 - b. The nature of the entity (i.e., superclass, subclass, regular, or weak entity): Subclass, Regular
 - c. Name(s) of the entity's superclass(es), if applicable: Businesses
 - d. Name(s) of its subclass(es), if applicable: N/A

- e. Its attributes: ID number (Plumb#), Name (Plumb_Name), Location (Plumb_Loc), Phone number (Plumb_Phone), Operating hours (Plumb_OpHours), Specialty (Plumb_Specialty), Accepted payment methods (Plumb_AccPmt), Description (Plumb_Descript)
 - f. Its relationship(s) with other entities, if applicable: Type of Business
26. Each Book has several Hotels. Each Hotel has an identification number, name, phone number, location, website, price range, booking restrictions, amenities, payment method accepted, historic area, description, and cancellation restrictions,
- a. Name of the entity: Hotels
 - b. The nature of the entity (i.e., superclass, subclass, regular, or weak entity): Subclass, Regular
 - c. Name(s) of the entity's superclass(es), if applicable: Book
 - d. Name(s) of its subclass(es), if applicable: N/A
 - e. Its attributes: ID number (Hotel#), Name (Hotel_Name), Phone number (Hotel_Phone), Location (Hotel_Loc), Website (Hotel_Web), Price range (Hotel_PriceRange), Booking restrictions (Hotel_BookRestrict), Amenities (Hotel_Amenities), Accepted payment methods (Hotel_AccPmt), Historic area (Hotel_HisArea), Description (Hotel_Descript), Cancellation restrictions (Hotel_CancelRestrict)
 - f. Its relationship(s) with other entities, if applicable: Is section of Book
27. Each Book has several Restaurants. Each Restaurant has an identification number, name, phone number, location, website, food specialty, may or may not accept reservations, and description.
- a. Name of the entity: Restaurants
 - b. The nature of the entity (i.e., superclass, subclass, regular, or weak entity): Subclass, Regular
 - c. Name(s) of the entity's superclass(es), if applicable: Book
 - d. Name(s) of its subclass(es), if applicable: N/A
 - e. Its attributes: ID number (Rest#), Name (Rest_Name), Phone number (Rest_Phone), Location (Rest_Loc), Website (Rest_Web), Food specialty (Rest_CuisineType), Reservation policy (Rest_Reserve), Description (Rest_Descript)
 - f. Its relationship(s) with other entities, if applicable: Is section of Book

III. Extended Entity Relation (EER) Diagram Development:

III.1 Constraints:

Using solely the documentation provided as the source of the data (i.e. Book titled “Glorious Savannah”), this EER design has **five** total constraints that are all **Disjoint (d) with Total**

Participation on Specialization. The entities with this constraint and the reasonings are listed as follows:

- Entity: Book
 - Disjoint: The instances of the **Book** type are members of maximum one subclass.
 - Total Participation: It is mandatory that every instance of the **Book** type be a member of at least one subclass.
- Entity: Attractions
 - Disjoint: The instances of the **Attractions** type are members of maximum one subclass.
 - Total Participation: It is mandatory that every instance of the **Attractions** type be a member of at least one subclass.
- Entity: Sports
 - Disjoint: The instances of the **Sports** type are members of maximum one subclass.
 - Total Participation: It is mandatory that every instance of the **Sports** type be a member of at least one subclass.
- Entity: Universities
 - Disjoint: The instances of the **Universities** type are members of maximum one subclass.
 - Total Participation: It is mandatory that every instance of the **Universities** type be a member of at least one subclass.
- Entity: Businesses
 - Disjoint: The instances of the **Businesses** type are members of maximum one subclass.
 - Total Participation: It is mandatory that every instance of the **Businesses** type be a member of at least one subclass.

III.2 EER Diagram:

IV. Conversion to Relations:

Step 8: Convert each specialization with m subclasses $\{S_1, S_2, \dots, S_m\}$ and superclass of C , where the attributes of C are $\{k, a_1, \dots, a_n\}$ and k is the primary key, into relation schemas using one of the following options:

- **Option 8-A:** Create a relation L for C with attributes $\text{Attr}(L) = \{k, a_1, \dots, a_n\}$ and $\text{PK}(L) = k$. Also create a relation L_i for each subclass S_i (for $1 \leq i \leq m$), with the attributes $\text{Atts}(L_i) = \{k\} \cup \{\text{attributes of } S_i\}$ and $\text{PK}(L_i) = k$.
 - **Entity/Superclass (C):** Book
 - Attributes of C ($k, a_1 \dots a_n$): ISBN, Title, Publisher, Edition#, PubYear, NumPgs
 - Primary Key (k): ISBN
 - **Final Relation (L):** Book(ISBN, Title, Publisher, Edition#, PubYear, NumPgs)
 - **Entity/Subclass of C (S_i)/Superclass (C):** Attractions
 - Attributes of S_i ($k, a_1 \dots a_n$): Att#
 - Primary Key Derived from C (k): ISBN

- Composite Key: ISBN, Att#
- **Final Relation (Li): Attractions(ISBN, Att#)**
 - **Entity/Subclass of C (Si): Emergency Agencies**
 - Attributes of Si (k, a1...an): Emergency#, Emergency_Name, Emergency_Phone
 - Primary Key Derived from C (k): Att#
 - Composite Key: ISBN, Att#, Emergency#
 - **Final Relation (Li): Emergency_Agencies (ISBN, Att#, Emergency#, Emergency_Name, Emergency_Phone)**
 - **Entity/Subclass of C (Si): Savannah Attractions**
 - Attributes of Si (k, a1...an): SavAtt#, SavAtt_Name, SavAtt_Loc, SavAtt_Phone, SavAtt_Web, SavAtt_Descript
 - Primary Key Derived from C (k): Att#
 - Composite Key: ISBN, Att#, SavAtt#
 - **Final Relation (Li): Sav_Att(ISBN, Att#, SavAtt#, SavAtt_Name, SavAtt_Loc, SavAtt_Phone, SavAtt_Web, SavAtt_Descript)**
 - **Entity/Subclass of C (Si): Tybee Attractions**
 - Attributes of Si (k, a1...an): TybAtt#, TybAtt_Name, TybAtt_Loc, TybAtt_Phone, TybAtt_Web, TybAtt_Descript
 - Primary Key Derived from C (k): Att#
 - Composite Key: ISBN, Att#, TybAtt#
 - **Final Relation (Li): Tyb_Att(ISBN, Att#, TybAtt#, TybAtt_Name, TybAtt_Loc, TybAtt_Phone, TybAtt_Web, TybAtt_Descript)**
 - **Entity/Subclass of C (Si): Historic Cemeteries**
 - Attributes of Si (k, a1...an): Cemetery#, Cemetery_Name, Cemetery_Loc, Cemetery_EstYear, Cemetery_Type, Cemetery_Descript
 - Primary Key Derived from C (k): Att#
 - Composite Key: ISBN, Att#, Cemetery#
 - **Final Relation (Li): Historic_Cemeteries(ISBN, Att#, Cemetery#, Cemetery_Name, Cemetery_Loc, Cemetery_EstYear, Cemetery_Type, Cemetery_Descript)**
 - **Entity/Subclass of C (Si): Historic Churches**
 - Attributes of Si (k, a1...an): Church#, Church_Name, Church_Phone, Church_Loc, Church_Denom, Church_Web, Church_Descript

- Primary Key Derived from C (k): Att#
 - Composite Key: ISBN, Att#, Church#
 - **Final Relation (Li): Historic_Churches(ISBN, Att#, Church#, Church_Name, Church_Phone, Church_Loc, Church_Denom, Church_Web, Church_Descript)**
- **Entity/Subclass of C (Si): Historic Squares**
 - Attributes of Si (k, a1...an): Square#, Square_Name, Square_Loc, Square_Designer, Square_DesignYear, Square_ArchStyle, Square_Descript
 - Primary Key Derived from C (k): Att#
 - Composite Key: ISBN, Att#, Square#
 - **Final Relation (Li): Historic_Squares(ISBN, Att#, Square#, Square_Name, Square_Loc, Square_Designer, Square_DesignYear, Square_ArchStyle, Square_Descript)**
- **Entity/Subclass of C (Si): Parks**
 - Attributes of Si (k, a1...an): Park#, Park_Name, Park_Loc, Park_Munic, Park_Descript
 - Primary Key Derived from C (k): Att#
 - Composite Key: ISBN, Att#, Park#
 - **Final Relation (Li): Parks(ISBN, Att#, Park#, Park_Name, Park_Loc, Park_Munic, Park_Descript)**
- **Entity/Subclass of C (Si): CalEvents**
 - Attributes of Si (k, a1...an): CalEvent#, CalEvent_Name, CalEvent_Phone, CalEvent_Loc, CalEvent_Web, CalEvent_StartDate, CalEvent_Descript
 - Primary Key Derived from C (k): Att#
 - Composite Key: ISBN, Att#, CalEvent#
 - **Final Relation (Li): CalEvents(ISBN, Att#, CalEvent#, CalEvent_Name, CalEvent_Phone, CalEvent_Loc, CalEvent_Web, CalEvent_StartDate, CalEvent_Descript)**
- **Entity/Subclass of C (Si): Art**
 - Attributes of Si (k, a1...an): Art#, Art_Name, Art_Phone, Art_Loc, Art_Web, Art_Type
 - Primary Key Derived from C (k): Att#
 - Composite Key: ISBN, Att#, Art#
 - **Final Relation (Li): Art(ISBN, Att#, Art#, Art_Name, Art_Phone, Art_Loc, Art_Web, Art_Type)**
- **Entity/Subclass of C (Si)/Superclass (C): Sports**

- Attributes of Si (k, a1...an): Sports#
- Primary Key Derived from C (k): Att#
- **Final Relation (Li): Sports(ISBN, Att#, Sports#)**
 - **Entity/Subclass of C (Si): Tennis Courts**
 - Attributes of Si (k, a1...an): Tennis#, Tennis_Name, Tennis_Phone, Tennis_Loc, Tennis_Web, Tennis_Descript
 - Primary Key Derived from C (k): Sports#
 - Composite Key: ISBN, Att#, Sports#, Tennis#
 - **Final Relation (Li): Tennis_Courts(ISBN, Att#, Sports#, Tennis#, Tennis_Name, Tennis_Phone, Tennis_Loc, Tennis_Web, Tennis_Descript)**
 - **Entity/Subclass of C (Si): Golf Courses**
 - Attributes of Si (k, a1...an): Golf#, Golf_Name, Golf_Phone, Golf_Loc, Golf_Web, Golf_Descript
 - Primary Key Derived from C (k): Sports#
 - Composite Key: ISBN, Att#, Sports#, Golf#
 - **Final Relation (Li): Golf_Courses(ISBN, Att#, Sports#, Golf#, Golf_Name, Golf_Phone, Golf_Loc, Golf_Web, Golf_Descript)**
 - **Entity/Subclass of C (Si): Boat Ramps**
 - Attributes of Si (k, a1...an): Ramp#, Ramp_Name, Ramp_Loc
 - Primary Key Derived from C (k): Sports#
 - Composite Key: ISBN, Att#, Sports#, Ramp#
 - **Final Relation (Li): Boat_Ramps(ISBN, Att#, Sports#, Ramp#, Ramp_Name, Ramp_Loc)**
 - **Entity/Subclass of C (Si): Fishing Piers**
 - Attributes of Si (k, a1...an): Pier#, Pier_Name, Pier_Loc
 - Primary Key Derived from C (k): Sports#
 - Composite Key: ISBN, Att#, Sports#, Pier#

- **Final Relation (Li): Fishing_Piers**(ISBN, Att#, Sports#, Pier#, Pier_Name, Pier_Loc)
- **Entity/Subclass of C (Si)/Superclass (C): Universities**
 - Attributes of Si (k, a1...an): Uni#, Uni_Name, Uni_Phone, Uni_Email, Uni_Loc, Uni_Web, Uni_Type, Uni_Descript
 - Primary Key Derived from C (k): Att#
 - Composite Key: ISBN, Att#, Uni#
 - **Final Relation (Li): Universities**(ISBN, Att#, Uni#, Uni_Name, Uni_Phone, Uni_Email, Uni_Loc, Uni_Web, Uni_Type, Uni_Descript)
 - **Entity/Subclass of C (Si): Uni_Colleges**
 - Attributes of Si (k, a1...an): UniCollege#, UniCollege_Name
 - Primary Key Derived from C (k): Uni#
 - Composite Key: ISBN, Att#, Uni#, Uni_College#
 - **Final Relation (Li): Uni_Colleges**(ISBN, Att#, Uni#, UniCollege#, UniCollege_Name)
 - **Entity/Subclass of C (Si): Uni_Programs**
(UniProg#, UniProg_Name)
 - Attributes of Si (k, a1...an): UniProg#, UniProg_Name
 - Primary Key Derived from C (k): Uni#
 - Composite Key: ISBN, Att#, Uni#, Uni_Prog#
 - **Final Relation (Li): Uni_Programs**(ISBN, Att#, Uni#, UniProg#, UniProg_Name)
 - **Entity/Subclass of C (Si): Uni_Depts**
 - Attributes of Si (k, a1...an): UniDept#, UniDept_Name, UniDept_Phone
 - Primary Key Derived from C (k): Uni#
 - Composite Key: ISBN, Att#, Uni#, UniDept#
 - **Final Relation (Li): Uni_Depts**(ISBN, Att#, Uni#, UniDept#, UniDept_Name, UniDept_Phone)
 - **Entity/Subclass of C (Si)/Superclass (C): Businesses**
 - Attributes of Si (k, a1...an): BusID

- Primary Key Derived from C (k): ISBN
- Composite Key: ISBN, BusID
- **Final Relation (Li): Businesses(ISBN, BusID)**
 - **Entity/Subclass of C (Si): Medical Establishments**
 - Attributes of Si (k, a1...an): Med#, MedEst_Name, MedEst_Specialty, MedEst_Descript, MedEst_Services, MedEst_Loc, MedEst_OpHours, MedEst_Web, MedEst_Phone
 - Primary Key Derived from C (k): BusID
 - Composite Key: ISBN, BusID, Med#
 - **Final Relation (Li): MedEsts(ISBN, BusID, Med#, MedEst_Name, MedEst_Specialty, MedEst_Descript, MedEst_Services, MedEst_Loc, MedEst_OpHours, MedEst_Web, MedEst_Phone)**
 - **Entity/Subclass of C (Si): Pest Control Companies**
 - Attributes of Si (k, a1...an): Pest#, Pest_Name, Pest_Loc, Pest_Phone, Pest_OpHours, Pest_Specialty, Pest_AccPmt, Pest_Descript
 - Primary Key Derived from C (k): BusID
 - Composite Key: ISBN, BusID, Pest#
 - **Final Relation (Li): PestControl(ISBN, BusID, Pest#, Pest_Name, Pest_Loc, Pest_Phone, Pest_OpHours, Pest_Specialty, Pest_AccPmt, Pest_Descript)**
 - **Entity/Subclass of C (Si): Plumbers**
 - Attributes of Si (k, a1...an): Plumb#, Plumb_Name, Plumb_Loc, Plumb_Phone, Plumb_OpHours, Plumb_Specialty, Plumb_AccPmt, Plumb_Descript
 - Primary Key Derived from C (k): BusID
 - Composite Key: ISBN, BusID, Plumb#
 - **Final Relation (Li): Plumbers(ISBN, BusID, Plumb#, Plumb_Name, Plumb_Loc, Plumb_Phone, Plumb_OpHours, Plumb_Specialty, Plumb_AccPmt, Plumb_Descript)**
- **Entity/Subclass of C (Si): Hotels**
 - Attributes of Si (k, a1...an): Hotel#, Hotel_Loc, Hotel_Name, Hotel_Phone, Hotel_Web, Hotel_HisArea, Hotel_BookRestrict, Hotel_PriceRange, Hotel_AccPmt, Hotel_CancelRestrict, Hotel_Amenities, Hotel_Descript
 - Primary Key Derived from C (k): ISBN
 - Composite Key: ISBN, Hotel#

- **Final Relation (Li): Hotels**(ISBN, Hotel#, Hotel_Loc, Hotel_Name, Hotel_Phone, Hotel_Web, Hotel_HisArea, Hotel_BookRestric, Hotel_PriceRange, Hotel_AccPmt, Hotel_CancelRestric, Hotel_Amenities, Hotel_Descript)
- **Entity/Subclass of C (Si): Restaurants**
 - Attributes of Si (k, a1...an): Rest#, Rest_Name, Rest_Phone, Rest_Loc, Rest_Web, Rest_CuisineType, Rest_Reserve, Rest_Descript, Rest_OpHours
 - Primary Key Derived from C (k): ISBN
 - Composite Key: ISBN, Rest#
 - **Final Relation (Li): Restaurants**(ISBN, Rest#, Rest_Loc, Rest_Name, Rest_Phone, Rest_Web, Rest_Reserve, Rest_CuisineType, Rest_Descript, Rest_OpHours)

Step 9: Mapping of a Lattice into relations:

Case A (Shared Subclass): The superclasses of the shared subclass have the same key, {k}. Create a relation, L for the shared subclass with attributes $A_{\text{tr}}(L) = \{k\} \cup \{\text{attributes of the shared subclass}\}$ and $PK(L) = \{k\}$

Case B (Category): The superclasses of the category have different keys. Create a relation, L, for the category with attributes $A_{\text{tr}}(L) = \{\text{key}\} \cup \{\text{all the attributes of the category subclass}\}$ and $PK(L) = \{\text{key}\}$; {key} is a newly created key attribute for the category and it is called a surrogate key. Add the surrogate key to every subclass's relation.

- **Since none of the entities have multiple superclasses, the EER model contains no lattice structures; as such this rule is not applicable.**

Step 1: For each regular entity type E, create a relation P that includes all the simple attributes of E. Include only the simple component attributes of composite attributes of E. Choose one of the key attributes of E as the primary key for P. If the chosen key of E is composite, the set of simple attributes that form it will together make the primary key of P.

- **Since Step 8, Option 8-A handled the conversion of the regular entities and their attributes in the diagram, applying Step 1 is redundant.**

Step 2: For each weak entity type W with owner entity type E, create a relation P, and include all simple attributes (or simple components of composite attributes) of W in P. In addition, include the primary key attribute(s) of the relation(s) that correspond to the owner entity type(s) to P. The primary key of P is the combination of the primary key(s) of the owner(s) and the partial key of the weak entity type W, if any.

1. Weak Entity (W): Doctors
 - a. Relation Name (P): Doctors
 - b. Simple Attributes: Doc#, Doc_Name, Doc_Degree

- c. Key Attribute: Med#
- d. Partial Key: Doc#
- e. Final Relation (P): **Doctors(Med#, Doc#, Doc_Name, Doc_Degree)**

Step 3: For each binary 1:1 relationship type R, identify the relations S and T that represent the entity types participating in R. Choose one of the relations—S, say—and include as foreign key in S the primary key of T. (It is better to choose an entity type with total participation in R in the role of S.) Include all the simple attributes and simple components of composite attributes of the 1:1 relationship type R as attributes of S.

- Since there is **no binary 1:1 relationship type R** in this diagram, this rule is not applicable.

Step 4: For each binary 1:N relationship type R, identify the relations S and T that represent the entity types participating in R (S is in the N-side and T is in 1-side). Include as foreign key in S the primary key of the relation T. Include also any simple attribute and simple components of composite attributes of type R as attributes of S.

- Since **Step 2** handles the weak relationship of **MedEsts to Doctors** entirely by incorporating **MedEst's** key into the **Doctor's** relation as part of the primary key, applying Step 4 is **redundant**.

Step 5: For each binary M:N relationship type R, identify the relations S and T that represent the entity types participating in R. Create a new relation P to represent R. Include the primary keys of S and T in P that collectively makes the primary key of P and they also act as foreign keys. Include also the simple attributes and simple components of composite attributes of type R as attributes of P.

- Since there is **no binary M:N relationship type R** in this diagram, this rule is not applicable.

Step 6: For each n-ary ($n > 2$) relationship type R, identify the relations $S_1, \dots, S_n$ correspond to the n entity types participating in R. If entity type E_i has the max participation of 1, then mark relation S_i . Create a new relation P to represent R. Include the primary keys of $S_1, \dots, S_n$ in P as foreign keys. Include also the simple attributes and simple components of composite attributes of type R as attributes of P. If there is a marked relation S_i , then the key of S_i in P is the primary key of P; otherwise, all the foreign keys in P collectively make the primary key of P.

- Since there is **no n-ary relationship type R** in this diagram, this rule is not applicable.

Step 7: For each multivalued attribute A that belongs to the entity type E or relationship type R (let S be the relation that represents E or R), create a new relation P that includes an attribute corresponding to A plus the primary key, K, of S (as the foreign key). Combination of A and K makes the primary key of relation P.

1. Multivalued Attribute (A): Doc_Degree
 - a. Relation Representing E/R (S): Doctors
 - b. Attribute(s) Corresponding to A: Degree_Name, Degree_Lvl
 - c. Primary Key of S (K): Med#, Doc#
 - d. Primary Key of Relation (Combo of A and K): Med#, Doc#, Degree_Name
 - e. **Final Relation (P): Doctor_Degree(Med#, Doc#, Degree_Name, Degree_Lvl)**

V. List of Attributes:

Attribute Name	Attribute Description	Example of Attribute
Art_Loc	Art Location	204 W St. Julian St.
Art_Name	Art Name	City Market Art Center
Art_Phone	Art Phone Number	912-232-4903
Art_Type	Art Type (Concert, Tea, etc)	Concert
Art_Web	Art Website	www.savannahcitymarket.com
Art#	Art Identification	ART001
Att#	Attraction Identification	A001
BusID	Business Identification	BUS001
CalEvent_Descript	Event Description	Savannah Philharmonic Chorus, Sara Peebles, Ian O'Brien
CalEvent_Loc	Event Location	Trustees Theater
CalEvent_Name	Calendar of Events Event Name	A Night in Vienna

Attribute Name	Attribute Description	Example of Attribute
CalEvent_Phone	Event Phone	912-525-5050
CalEvent_StartDate	Event Start Date	2/1/2013
CalEvent_Web	Event Website	www.savannahboxoffice.com
CalEvent#	Calendar of Events Identification	CA001
Cemetery_Descript	Cemetery Description	Conrad Alken, Johnny Mercer, oaks, statuary, headstones
Cemetery_EstYear	Cemetery Year Established	1750
Cemetery_Loc	Cemetery Location	Wheaton St to Bonaventure Rd.
Cemetery_Name	Cemetery Name	Bonaventure Cemetery
Cemetery_Type	Cemetery Type	Historic
Cemetery#	Cemetery Identification	HCE01
Church_Denom	Church Denomination	Baptist
Church_Descript	Church Description	Organized in 1800, chartered in 1801 by Gov. Josiah Tattnall Jr, moved in 1833
Church_Loc	Church Location	233 Bull St.
Church_Name	Church Name	First Baptist Church
Church_Phone	Church Phone	912-234-2671

Attribute Name	Attribute Description	Example of Attribute
Church_Web	Church Website	www.christchurchsavannah.org
Church#	Church Identification	HCH01
Cuisine_Type	Restaurant Cuisine Offered	Fast Food, Seafood
Degree_Lvl	Level of College Degree	Bachelor's (BS)
Degree_Name	Name of College Major on Degree	Biological and Biomedical Sciences
Doc_Degree	Doctor's Degree(s)	M.D.
Doc_Name	Doctor's Name	Robert W. Beard
Doc#	Doctor Identification Number	DR001
Edition	Book Edition Number	2nd
Emergency_Name	Emergency Name	Chatham Emergency Management Agency
Emergency_Phone	Emergency Phone Number	912-201-4500
Emergency#	Emergency Identification	E1000
Golf_Descript	Golf Course Description	27 holes, par 72
Golf_Loc	Golf Course Location	One Shorty Cooper Dr.
Golf_Name	Golf Course Name	Bacon Park Golf Course
Golf_Phone	Golf Course Phone Number	912-354-2625

Attribute Name	Attribute Description	Example of Attribute
Golf_Web	Golf Course Website	www.baconparkgolf.com
Golf#	Golf Course Identification	GC001
Hotel_AccPmt	Hotel Payment Accepted	Visa, Mastercard, American Express
Hotel_Amenities	Hotel Amenities Available	Heated outdoor pool, whirlpool, exercise room
Hotel_BookRestrict	Restrictions on Booking Hotel	2-night minimum on Fri–Sat; 72-hour cancellation or first night charged
Hotel_CancelRestrict	Hotel Restrictions on Cancellation	Cancellation fee imposed
Hotel_Descript	Hotel Description	Boutique B&B with antique décor
Hotel_HisArea	Hotel Historic Area	Savannah
Hotel_Loc	Hotel Location	14 E Oglethorpe Ave, Savannah, GA 31401
Hotel_Name	Hotel Name	Ballastone Inn
Hotel_Phone	Hotel Phone Number	(912) 555-0142
Hotel_PriceRange	Hotel Price Range	\$\$\$ (200–350 per night)
Hotel_Web	Hotel Website	www.hotelname.com
Hotel#	Hotel Identification	H0001
ISBN	International Serial Book Number	1234-53-456-78

Attribute Name	Attribute Description	Example of Attribute
Med_Descript	Medical Establishment Description	Comprehensive Ophthalmology & Emergency Services
Med_Loc	Medical Establishment Location	5354 Reynold's St., Suite 225, Candler Professional Building, Savannah, GA 31405
Med_Name	Medical Establishment Name	The Neurological Institute of Savannah & Center for Spine
Med_OpHours	Medical Establish Operation Hours	9:00-5:00 Mon, Tues, Thurs, Fri; 9:00-1:00 Wed
Med_Phone	Medical Establishment Phone Number	912-352-3120
Med_Services	Medical Establishment Services	Cataract Surgery, LASIK & PRK
Med_Specialty	Medical Specialty	Ophthalmology (Eye)
Med_Web	Medical Establishment Website	www.myvisualbest.com
Med#	Medical Establishment Identification	12
NumPgs	Total number of pages in a book	72
Park_Descript	Park Description	1.2 acres, playground, athletic field, picnic area, spray pool
Park_Loc	Park Location	2100 Texas/Ohio Ave
Park_Munic	Park Municipality	City
Park_Name	Park Name	Avondale Park

Attribute Name	Attribute Description	Example of Attribute
Park#	Park Identification	P1000
Pest_AccPmt	Pest Control Payment Accepted	Discover
Pest_Descript	Pest Control Description	Certified and Insured for Termite Inspections, Lower Rates, Superior Service
Pest_Loc	Pest Control Location	Savannah
Pest_Name	Pest Control Name	Cox Termite and Pest Control
Pest_OpHours	Pest Control Hours of Operation	M-Su 24hr
Pest_Phone	Pest Control Phone Number	912-651-9000
Pest_Specialty	Pest Control Specialty	Roaches, Ants, Termites
Pest#	Pest Control Identification	PC005
Pier_Loc	Fishing Pier Location	Island Expy
Pier_Name	Fishing Pier Name	Frank W. Spencer
Pier#	Fishing Pier Identification	FP001
Plumb_AccPmt	Plumber Company Payment Accepted	Mastercard
Plumb_Descript	Plumber Company Description	Guaranteed workmanship, on-time and rapid response
Plumb_Loc	Plumber Company Location	5019 Ogeechee Rd., Savannah, GA 31312

Attribute Name	Attribute Description	Example of Attribute
Plumb_Name	Plumber Name	Thompson & Thompson Service Group
Plumb_OpHours	Plumber Company Hours of Operation	M-Su 24hr
Plumb_Phone	Plumber Company Phone	912-373-8511
Plumb_Specialty	Plumber Company Specialty	Water Heaters, Sump Pumps
Plumb#	Plumber Company Identification	PL002
Pub	Publisher of book	Focus
PubYear	Year of Publication	2020
Ramp_Loc	Boat Ramp Location	6851 Ogeechee Rd
Ramp_Name	Boat Ramp Name	Kings Ferry
Ramp#	Boat Ramp Identification	BR010
Rest_Descript	Restaurant Description	Casual waterfront spot known for shrimp and grits
Rest_Loc	Restaurant Location	412 W River St, Savannah, GA 31401
Rest_Name	Restaurant Name	River Street Seafood Co.
Rest_OpHours	Restaurant Hours of Operation	Mon–Thu 11:00–21:00; Fri–Sat 11:00–23:00; Sun 10:00–15:00 (brunch)
Rest_Phone	Restaurant Name	(912) 555-0198

Attribute Name	Attribute Description	Example of Attribute
Rest_Reserve	Restaurant Accepts Reservations?	Yes, via OpenTable; walk-ins accepted
Rest_Web	Restaurant Website	www.restaurantname.com
Rest#	Restaurant Identification	REST0001
Sav_Att_Descript	Savannah Attraction Description	11 ft bronze monument of a black family embracing in shackles
Sav_Att_Loc	Savannah Attraction Location	Rousakis Plaza/River St.
Sav_Att_Name	Savannah Attraction Name	African-American Monument
Sav_Att_Phone	Savannah Attraction Phone Number	912-233-6854
Sav_Att_Web	Savannah Attraction Website	www.andrewlowhouse.com
Sav_Att#	Savannah Attraction Identification	SA100
Sports#	Sports Identification	S001
Square_Descript	Square Description	Only a small portion remains
Square_DesignYear	Square Designed Year	1851
Square_Inspiration	Square Inspiration	John C. Calhoun
Square_Loc	Square Location	Abercorn St between Taylor and Gordon Sts
Square_Name	Square Name	Calhoun Square
Square#	Square Identification	HS100

Attribute Name	Attribute Description	Example of Attribute
Tennis_Descript	Tennis Court Description	16 courts, all lit
Tennis_Loc	Tennis Court Location	6262 Skidaway Rd.
Tennis_Name	Tennis Court Name	Bacon Parl
Tennis_Phone	Tennis Court Phone	912-351-3850
Tennis#	Tennis Court Identification	TC001
Title	Title of book	Glorious Savannah
Tyb_Att_Descript	Tybee Island Attraction Description	One of last coastal artillery batteries built along East Coast
Tyb_Att_Loc	Tybee Island Attraction Location	US Highway 80
Tyb_Att_Name	Tybee Island Attraction Name	Tybee Island Historical Society - Fort Screven
Tyb_Att_Phone	Tybee Island Attraction Phone	912-786-5801
Tyb_Att_Web	Tybee Island Attraction Website	www.tybeelighthouse.org
Tyb_Att#	Tybee Island Attraction Identification	TB100
UniCollege#	University College Identification	UC0001
UniCollege_Name	Name of University College	Business Administration
Uni_Descript	University Description	Catholic, liberal arts, chartered in 1889
Uni_Email	University Email Address	savannah@saintleo.edu

Attribute Name	Attribute Description	Example of Attribute
Uni_Loc	University Location	1332 Southern Dr
Uni_Name	University Name	Georgia Southern
Uni_Phone	University Phone Number	912-478-4636
UniProg#	University Program Identification Number	UP0001
UniProg_Name	University Program Name	Science and Technology
Uni_Type	University Type (2/4 year, etc)	2 year
Uni_Web	University Website	www.georgiasouthern.edu
Uni#	University Identification	UN002
UniDept#	University Business Departments	UD0201
UniDept_Name	University Business Department Names	Admissions
UniDept_Phone	University Business Department Phone Numbers	912-478-5391

VI. Final Set of Relations:

- Art(**ISBN**, **Att#**, **Art#**, Art_Name, Art_Phone, Art_Loc, Art_Web, Art_Type)
- Attractions(**ISBN**, **Att#**)
- Boat_Ramps(**ISBN**, **Att#**, **Sports#**, **Ramp#**, Ramp_Name, Ramp_Loc)
- Book(**ISBN**, Title, Publisher, Edition#, PubYear, NumPgs)
- Businesses(**ISBN**, **BusID**)
- CalEvents(**ISBN**, **Att#**, **CalEvent#**, CalEvent_Name, CalEvent_Phone, CalEvent_Loc, CalEvent_Web, CalEvent_StartDate, CalEvent_Descript)
- Doc_Degree(**Med#**, **Doc#**, **Degree_Name**, Degree_Lvl)
- Doctors(**Med#**, **Doc#**, Doc_Name, Doc_Degree)

- Emergency_Agencies(**ISBN, Att#, Emergency#**, Emergency_Name, Emergency_Phone)
- Fishing_Piers(**ISBN, Att#, Sports#, Pier#**, Pier_Name, Pier_Loc)
- Golf_Courses(**ISBN, Att#, Sports#, Golf#**, Golf_Name, Golf_Phone, Golf_Loc, Golf_Web, Golf_Descript)
- Historic_Cemeteries(**ISBN, Att#, Cemetery#**, Cemetery_Name, Cemetery_Loc, Cemetery_EstYear, Cemetery_Type, Cemetery_Descript)
- Historic_Churches(**ISBN, Att#, Church#**, Church_Name, Church_Phone, Church_Loc, Church_Denom, Church_Web, Church_Descript)
- Historic_Squares(**ISBN, Att#, Square#**, Square_Name, Square_Loc, Square_Designer, Square_DesignYear, Square_ArchStyle, Square_Descript)
- Hotels(**ISBN, Hotel#**, Hotel_Loc, Hotel_Name, Hotel_Phone, Hotel_Web, Hotel_HisArea, Hotel_BookRestric, Hotel_PriceRange, Hotel_AccPmt, Hotel_CancelRestric, Hotel_Amenities, Hotel_Descript)
- MedEsts(**ISBN, BusID, Med#**, MedEst_Name, MedEst_Specialty, MedEst_Descript, MedEst_Services, MedEst_Loc, MedEst_OpHours, MedEst_Web, MedEst_Phone)
- Parks(**ISBN, Att#, Park#**, Park_Name, Park_Loc, Park_Munic, Park_Descript)
- PestControl(**ISBN, BusID, Pest#**, Pest_Name, Pest_Loc, Pest_Phone, Pest_OpHours, Pest_Specialty, Pest_AccPmt, Pest_Descript)
- Plumbers(**ISBN, BusID, Plumb#**, Plumb_Name, Plumb_Loc, Plumb_Phone, Plumb_OpHours, Plumb_Specialty, Plumb_AccPmt, Plumb_Descript)
- Restaurants(**ISBN, Rest#**, Rest_Loc, Rest_Name, Rest_Phone, Rest_Web, Rest_Reserve, Rest_CuisineType, Rest_Descript, Rest_OpHours)
- Sav_Att(**ISBN, Att#, SavAtt#**, SavAtt_Name, SavAtt_Loc, SavAtt_Phone, SavAtt_Web, SavAtt_Descript)
- Sports(**ISBN, Att#, Sports#**)
- Tennis_Courts(**ISBN, Att#, Sports#, Tennis#**, Tennis_Name, Tennis_Phone, Tennis_Loc, Tennis_Web, Tennis_Descript)
- Tyb_Att(**ISBN, Att#, TybAtt#**, TybAtt_Name, TybAtt_Loc, TybAtt_Phone, TybAtt_Web, TybAtt_Descript)
- Uni_Colleges(**ISBN, Att#, Uni#, UniCollege#**, UniCollege_Name)
- Uni_Depts(**ISBN, Att#, Uni#, UniDept#**, UniDept_Name, UniDept_Phone)
- Uni_Programs(**ISBN, Att#, Uni#, UniProg#**, UniProg_Name)
- Universities(**ISBN, Att#, Uni#**, Uni_Name, Uni_Phone, Uni_Email, Uni_Loc, Uni_Web, Uni_Type, Uni_Descript)